

Aryaroop Bose - PES1UG24CS087 - SE Lab 04

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Features / Bug Fixes

Scenario 23: Dots and Boxes · **Student:** PES1UG24CS087 · **Repo:** 23_dots_and_boxes

Bugs Fixed

1. **Broken board rendering** (in `board.py` - `display()`) — the old code used `".".join(...)`, which dropped the outer edge dots and produced misaligned rows (7-char horizontal rows vs 9-char vertical rows). Fixed to `"." + " ".join(...) + "."`, so every row is aligned and all dots are shown.
2. **No box ownership / unauditable score** (in `board.py`, `game.py`) — every claimed box displayed a generic X, so scores couldn't be traced to the board. Added an owner map `((row, col) → player)` and the board now marks claimed boxes 1/2 per player.
3. **Unvalidated line drawing** (in `board.py` - `add_line()`) — negative indices (e.g. `H -1 0`) silently corrupted the last row and out-of-range input raised a bare `IndexError`. `add_line()` now validates orientation, type, bounds, and duplicates, raising `ValueError` with a clear message, and returns the newly completed boxes.
4. **Crash on EOF / Ctrl-C** (in `game.py` - `run()`) — `input()` was unguarded, so piped input ending or Ctrl-D/C crashed with a traceback. Now handled with a clean abort that leaves board and score untouched.
5. **Fragmented input validation** (in `game.py`, `rules.py`) — parsing was inline with misleading messages. Centralized into `rules.parse_move()`, which returns specific errors for bad format, bad orientation, non-integers, out-of-range coordinates, and already-drawn lines; invalid input never mutates board or score, including moves attempted after game-over.

Features Added

1. **Configurable board size** (in `main.py`, `game.py`, `board.py`): `python main.py --rows 3 --cols 3` (1–9 supported); line totals and rendering adapt automatically.

2. **Undo** (in `game.py`, `board.py`): U command reverts the last move via move history, rolling back lines, box claims, scores, and turn (including double-box claims).
3. **Game stats & UX** (in `game.py`, `board.py`): lines-left counter in the header, move count and winner summary at game end, Q quit command, and testable `apply_move()` / `undo_last()` logic separated from I/O.

Automated tests (in `test_game.py`, new file): 12 unittest cases (stdlib only): valid horizontal/vertical moves, repeated/invalid moves, box completion with extra turn + ownership, double-box completion, end-of-game, malformed-input parsing, undo, and `add_line` validation.
Result: **12/12 pass**.